Notes: Berry and Haile (Handbook, 2021)

Foundations of Demand Estimation

Contents

1 Big picture

- **Question.** What makes demand estimation different from ordinary regression, and what data variation is needed to estimate demand elasticities and counterfactual responses credibly?
- **Object of interest.** The target is not a reduced-form average effect. It is the demand system: how quantities or market shares change when prices or product characteristics change while all demand shocks and other relevant market conditions are held fixed.
- **Why demand matters.** Demand elasticities are required for markups, merger simulation, tax pass-through, product introduction, school choice, health insurance, welfare, and almost any quantitative counterfactual involving market power.
- **First fundamental challenge.** Prices are endogenous because equilibrium prices respond to demand shocks and cost shocks. This is the classic simultaneity problem.
- **Second fundamental challenge.** In differentiated-product markets, demand for product j depends not only on its own unobservable, but also on demand shocks for all related products. This makes demand a system with multiple structural errors, not a scalar-error regression.
- **Important warning.** Even randomized prices are not enough, in general, to identify the demand curve. Randomization can break correlation between prices and demand shocks, but it does not hold the vector of demand shocks fixed.
- **Common tools fall short.** Controls, fixed effects, control functions, and LATE-style treatment-effect estimands often do not recover the elasticities or equilibrium counterfactual objects that motivate demand estimation.
- **Core solution.** The modern IO strategy is an **index–inversion–instruments** strategy: impose a useful index structure, invert the demand system to recover product-market demand shocks, and use instruments to discipline the endogenous variables.
- **Canonical empirical model.** The chapter explains random-coefficient discrete choice demand, especially the Berry–Levinsohn–Pakes (BLP) framework, because it balances tractability with flexible substitution patterns.
- **Instrument lesson.** With market-level data, instruments are needed not only for prices but also for quantities or shares. Cost shifters help with prices; BLP instruments and related demand-side shifters help with shares and markups.
- **Micro-data lesson.** Consumer-level choices and demographics provide within-market variation, where product-market demand shocks are fixed. This can replace some instruments and greatly improve identification of heterogeneity and substitution.
- **Main takeaway.** Demand estimation is hard not because IO uses complicated models for their own sake, but because the economic object requires holding fixed latent demand shocks in a multi-good equilibrium system.

2 Data and measurement: key objects

This is a methodological survey rather than an empirical paper. Thus, the "data" section should be read as a map of the data environments and measurement objects used in demand estimation.

2.1 Market-level data

In the market-level setting, the researcher observes markets $t = 1, \dots, T$, products $j = 1, \dots, J_t$, and aggregate outcomes:

- product prices p_{jt};
- product characteristics x_{jt};
- market shares $\tilde{s}_{jt}$, usually sales divided by market size;
- distributions of consumer demographics in each market, such as income or family size;
- possible instruments, such as cost shifters, rival characteristics, or neighboring-market demographics.

Interpretation. Market-level data are common in IO because firms and regulators often observe sales, prices, and characteristics by product and market, even when individual-level choice records are unavailable.

2.2 Micro data

Micro data match individual consumer characteristics d_{it} to individual choices q_{it}. Market-level data reveal only marginal distributions of demographics and choices, while micro data reveal their joint distribution.

Interpretation. The benefit is that one can compare consumers within the same market. Product-market demand shocks are fixed inside a market, so within-market variation in demographics and choices helps identify preference heterogeneity without relying entirely on cross-market instruments.

2.3 Consumer panels

Consumer panels observe the same consumer on multiple choice occasions. Examples include repeat grocery trips, repeated health-plan choices, or car purchases by the same household over time.

Interpretation. Panels add information about substitution. If the same consumer faces changed choice sets or prices over time, the model can better distinguish persistent tastes from product-level shocks.

2.4 Ranked choice data

Ranked choice data reveal a consumer's first choice, second choice, and possibly longer ranking. This can arise in conjoint surveys or strategy-proof school choice mechanisms.

Interpretation. Ranked choices are especially powerful because first-to-second-choice relationships directly reveal which products are close substitutes.

2.5 Hybrid data

Many applications combine aggregate market shares with limited micro data. Petrin-style applications, for example, combine market-level auto shares with survey data on household choices.

Interpretation. Hybrid estimation typically combines aggregate share moments with micro moments, such as the mean income of consumers choosing a product type.

2.6 Demand shocks

The key unobservable is a product-market demand shock, often denoted ξ_{jt}. It represents demand-relevant product-market factors observed by consumers and firms but not by the econometrician.

Interpretation. These shocks are the source of price endogeneity and the reason observed quantity-price movements are not automatically demand curves.

2.7 Elasticities and counterfactuals

The economic output is often an elasticity matrix,

$$\frac{\partial s_{jt}}{\partial p_{kt}}, \qquad j, k = 1, \ldots, J_t,$$

or a counterfactual equilibrium outcome under a tax, merger, tariff, new product, or different choice rule.

Interpretation. A differentiated-products demand system has many own- and cross-price elasticities. Estimation must therefore impose enough structure to be feasible while preserving realistic substitution patterns.

3 Models and empirical specifications

3.1 Single-good simultaneous demand and supply

The chapter begins with the simplest simultaneous-equations example:

$$Q = D(X, P, U), \tag{1}$$

$$P = C(W, Q, V). \tag{2}$$

Here Q is quantity, P is price, X is an observed demand shifter, W is an observed cost shifter, U is an unobserved demand shifter, and V is an unobserved cost shifter.

Interpretation. If U did not exist, the observed relationship between (X, P) and Q would be demand. Because U exists and affects equilibrium prices, prices are endogenous. Cost shifters W excluded from demand can serve as instruments.

3.2 Multi-good demand is not regression

With J related goods, demand for good j takes the form

$$Q_j = D_j(X, P, U), \tag{3}$$

where

$$X = (X_1, \ldots, X_J), \qquad P = (P_1, \ldots, P_J), \qquad U = (U_1, \ldots, U_J).$$

Interpretation. The demand equation for good j has a whole vector of structural errors on the right-hand side. This is why a standard IV regression logic with one scalar error is not enough.

3.3 Why randomized prices do not solve everything

Suppose a researcher randomly assigns the entire price vector P_t across markets. Then prices can be independent of demand shocks, but observed changes in demand integrate over different realizations of U_t.

Interpretation. Demand elasticities require varying prices while holding U_t fixed. A randomized price experiment usually identifies an average response over demand shocks, not the slope of demand at a known point.

3.4 Scalar-index special case

One way to make randomized prices sufficient is to impose a scalar-index restriction:

$$D_j(X, P, U) = D_j(X, P, E_j(U)), \tag{4}$$

where $E_j(U)$ is a scalar index and D_j is monotone in that index.

Interpretation. This collapses many demand shocks into one scalar object. It can make quantile or IV logic applicable, but it is a strong functional-form restriction and is not implied by many common differentiated-products demand models.

3.5 LATE and tax pass-through

For a single good with inverse supply $Q = S(W, P, V)$, the effect of an excise tax τ on equilibrium price depends on relative slopes:

$$\frac{\partial P^*(X, W, U, V, \tau)}{\partial \tau} = \frac{\partial S(W, P, V)/\partial P}{\partial S(W, P, V)/\partial P + \partial D(X, P, U)/\partial P}. \tag{5}$$

Interpretation. A local average treatment effect of price on quantity does not generally identify this ratio. Counterfactual equilibrium objects combine demand and supply slopes at the same point and under the same latent state.

3.6 Random utility discrete choice

In a random utility model, consumer i chooses the good with highest utility:

$$q_{ij} = 1\{u_{ij} \geq u_{ik} \ \forall k \in \{0, 1, \ldots, J_i\}\}. \tag{6}$$

Choice probabilities are

$$s_{ij} = \mathbb{E}[q_{ij} \mid J_i, \chi_i] = \int_{A_{ij}} dF_u(u_{i0}, u_{i1}, \ldots, u_{iJ_i} \mid J_i, \chi_i), \tag{7}$$

where A_{ij} is the region of utility space in which good j is preferred to all alternatives.

Interpretation. The model turns unobserved preference heterogeneity into probabilistic choice shares. The outside good is essential because without it total inside demand would be mechanically fixed.

3.7 Canonical random-coefficient utility

The canonical differentiated-products utility is

$$u_{ijt} = x_{jt}\beta_{it} - \alpha_{it}p_{jt} + \xi_{jt} + \varepsilon_{ijt}, \tag{8}$$

with outside option

$$u_{i0t} = \varepsilon_{i0t}. \tag{9}$$

A typical random-coefficient specification is

$$\beta_{it}^{(k)} = \beta_0^{(k)} + \sum_{\ell=1}^{L} \beta_d^{(\ell,k)} d_{i\ell t} + \beta_\nu^{(k)} \nu_{it}^{(k)}. \tag{10}$$

Interpretation. Consumers differ in tastes for product characteristics because they have different observed demographics d_{it} and different latent tastes ν_{it}. This produces richer substitution patterns than a simple logit model.

3.8 Why random coefficients matter

Without random coefficients, utility can be written as

$$u_{ijt} = x_{jt}\beta_0 - \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt}, \tag{11}$$
$$\delta_{jt} = x_{jt}\beta_0 - \alpha p_{jt} + \xi_{jt}, \tag{12}$$
$$u_{ijt} = \delta_{jt} + \varepsilon_{ijt}. \tag{13}$$

Interpretation. If the only product-specific object is mean utility δ_{jt}, products with similar market shares tend to have similar elasticities and cross-elasticities. Random coefficients let similar products be close substitutes because consumers with a high taste for one characteristic also prefer products sharing that characteristic.

3.9 BLP market-level model

The simplified BLP utility is

$$u_{ijt} = x_{jt}\beta_{it} - \alpha_0 p_{jt} + \xi_{jt} + \varepsilon_{ijt}. \tag{14}$$

It is rewritten as

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt}, \tag{15}$$
$$\delta_{jt} = x_{jt}\beta_0 - \alpha_0 p_{jt} + \xi_{jt}, \tag{16}$$
$$\mu_{ijt} = \sum_{k=1}^{K} x_{jt}^{(k)} \left(\sum_{\ell=1}^{L} \beta_d^{(\ell,k)} d_{i\ell t} + \beta_\nu^{(k)} \nu_{it}^{(k)} \right). \tag{17}$$

With type-I extreme-value ε_{ijt}, model-implied shares are

$$\sigma_j(\delta_t, x_t, \beta_d, \beta_\nu, J_t) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{J_t} \exp(\delta_{kt} + \mu_{ikt})} \, dF_\mu(\mu_{it} \mid x_t, \beta_d, \beta_\nu, J_t). \tag{18}$$

Interpretation. The key BLP insight is invertibility: for fixed nonlinear parameters and observed positive shares, there is a unique vector δ_t that makes predicted shares equal observed shares.

3.10 BLP GMM estimator

Partition parameters as

$$\theta_1 = (\alpha_0, \beta_0), \qquad \theta_2 = (\beta_d, \beta_\nu), \qquad \theta = (\theta_1, \theta_2).$$

The BLP estimator solves

$$\min_{\theta} \; g(\xi(\theta))' \Omega g(\xi(\theta)), \tag{19}$$

where

$$g(\xi(\theta)) = \frac{1}{N} \sum_{j,t} \xi_{jt}(\theta) z_{jt}, \tag{20}$$

$$\xi_{jt}(\theta) = \delta_{jt}(\theta_2) - x_{jt}\beta + \alpha p_{jt}, \tag{21}$$

$$\tilde{s}_{jt} = \sigma_j(\delta_t, x_t, \theta_2, J_t). \tag{22}$$

Interpretation. For each trial θ_2, the model is inverted to recover δ_t. The implied ξ_{jt} should be orthogonal to valid instruments z_{jt}. The GMM objective chooses parameters that make this orthogonality as close as possible in sample.

3.11 Adding a supply side

A common supply-side step recovers marginal cost by inverting firm first-order conditions:

$$mc_{jt} = \psi_j(s_t, p_t), \tag{23}$$

where ψ_j is known once the oligopoly model and demand slopes are known. A simple marginal cost function is

$$mc_{jt} = w_{jt}\gamma_0 + \gamma_1 q_{jt} + \omega_{jt}. \tag{24}$$

This gives supply moments

$$\mathbb{E}[\omega_{jt}(\theta, \gamma)\tilde{z}_{jt}] = 0. \tag{25}$$

Interpretation. The supply side can sharpen demand estimates because supply moments involve demand parameters. It can also test alternative assumptions about conduct when different conduct models imply different markup patterns.

4 Identification and instruments

4.1 Core identification logic

The chapter's central identification message is:

demand shocks create endogeneity ⇒ invert demand ⇒ use instruments for endogenous prices and quantities.

Interpretation. The inversion step transforms a system with many structural errors in each demand equation into equations with one recovered demand shock per product-market observation.

4.2 Why controls and fixed effects are limited

Adding controls or fixed effects can help only if they absorb all product-market demand shocks relevant for demand. This is usually implausible, especially when demand shocks vary by product and market.

Interpretation. Fixed effects are not a substitute for identifying price variation unless the remaining unobservables are irrelevant for demand or orthogonal to prices.

4.3 Why control functions are limited

Control-function approaches often rely on a triangular structure,

$$T = F(Z, E_1), \tag{26}$$

$$Y = G(T, E_2). \tag{27}$$

But in a demand-supply system, the reduced form for price is more like

$$P = R(X, W, U, V), \tag{28}$$

which depends on both demand and cost shocks.

Interpretation. A single control variate cannot generally hold fixed all structural shocks entering demand and price. The problem becomes more severe with many goods.

4.4 Market-level nonparametric demand

Berry and Haile's nonparametric market-level model writes demand as

$$s_{jt} = \sigma_j(x_t, p_t, \xi_t), \qquad j = 1, \ldots, J. \tag{29}$$

Partition $x_t = (x_t^{(1)}, x_t^{(2)})$ and impose the index restriction

$$\delta_{jt} = x_{jt}^{(1)} + \xi_{jt}. \tag{30}$$

Interpretation. The index says that one exogenous characteristic and the unobserved demand shock enter through a common scalar quality index for each product. This is the nonparametric analog of a mean-utility index.

4.5 Inversion under connected substitutes

Under a connected-substitutes condition, demand can be inverted:

$$\delta_{jt} = \sigma_j^{-1}(s_t; p_t), \qquad j = 1, \ldots, J. \tag{31}$$

Thus

$$x_{jt}^{(1)} = \sigma_j^{-1}(s_t; p_t) - \xi_{jt}. \tag{32}$$

Interpretation. This looks like a nonparametric IV equation, but with an exogenous product characteristic on the left and the endogenous share and price vectors on the right.

4.6 Instrument assumptions

The nonparametric identification result requires two familiar IV conditions:

- **Exclusion / mean independence:** $\mathbb{E}[\xi_{jt} \mid z_t, x_t^{(1)}] = 0$.
- **Relevance / completeness:** instruments must generate enough exogenous variation in (s_t, p_t) to identify the inverse demand functions.

Interpretation. Completeness is the nonparametric analog of a rank condition. It is the formal statement that the instruments move the endogenous objects in enough independent ways.

4.7 Why market-level data need instruments for $2J$ objects

The inverse demand function depends on $(s_t, p_t) \in \mathbb{R}^{2J}$. Therefore, in the fully nonparametric market-level model, one needs instruments that shift both prices and market shares.

Interpretation. Cost shifters can move prices, but they often move shares only through prices. To learn demand at fixed prices, the researcher also needs demand-side variation that shifts shares.

4.8 Cost shifters and Hausman instruments

Cost shifters include input prices, taxes, tariffs, distribution costs, or proxies for marginal cost. Hausman instruments use prices of the same product in other geographic markets as proxies for common cost shocks.

Interpretation. Cost instruments are valid only if they are independent of demand shocks. Hausman instruments require especially careful scrutiny because prices in other markets may also reflect correlated demand shocks.

4.9 BLP instruments

BLP instruments use exogenous characteristics of competing products, such as rival product characteristics or functions of the distribution of products in characteristic space.

Interpretation. Rival characteristics affect product j's market share and markup through substitution and competition. They are useful only if product characteristics are plausibly exogenous to demand shocks, or if the researcher has valid excluded variants of them.

4.10 Waldfogel-Fan instruments

Waldfogel-style instruments use demographic characteristics of neighboring or related markets. Fan-style logic applies when firms compete in partially overlapping service areas.

Interpretation. Neighboring consumers can affect the set of products and prices offered to a market, especially under zone pricing or shared service areas. These instruments are not the same as Hausman instruments: they operate through market demographics and markup incentives rather than common costs.

4.11 Exogenous market structure

Entry, exit, mergers, spinoffs, or changes in common ownership can shift competition and markups if they are exogenous to demand shocks.

Interpretation. Market-structure shifts can act as instruments because they change the pricing environment while not directly entering consumer utility, under the maintained exogeneity assumption.

4.12 Optimal instruments

Given a conditional moment $\mathbb{E}[\xi_{jt}(\theta_0) \mid z_{jt}] = 0$, Chamberlain's efficient instruments take the form

$$z_{jt}^* = \Psi_{jt}^{-1}\mathbb{E}\left[\left.\frac{\partial \xi_{jt}(\theta_0)}{\partial \theta}\right| z_{jt}\right], \tag{33}$$

where

$$\Psi_{jt} = \mathbb{E}[\xi_{jt}(\theta_0)^2 \mid z_{jt}]. \tag{34}$$

Interpretation. The best instruments are functions of the exogenous variables that predict how the structural residual changes with the parameters. They are infeasible but can be approximated; good approximations can greatly improve precision.

4.13 Micro data and reduced IV requirements

With micro data, conditional choice probabilities can vary with individual characteristics within a market:

$$s_{ijt} = \sigma_j(d_{it}, y_{it}, x_t, p_t, \xi_t). \tag{35}$$

Within a market, p_t, x_t, and ξ_t are fixed, while d_{it} varies across individuals.

Interpretation. Micro data can replace instruments for quantities because variation in demographics moves individual choice probabilities while holding market-level demand shocks fixed. However, instruments for prices are still generally required.

4.14 Nonparametric micro-data identification

Berry and Haile's micro-data argument uses an index of the form

$$\gamma_{jt} = g_j(d_{it}) + \xi_{jt}, \tag{36}$$

and exploits within-market variation in d_{it} plus cross-market variation in (p_t, ξ_t). A key support condition is the existence of a common choice-probability vector s^* reached in every market by some consumer type.

Interpretation. Once a common s^* is available, the inverse-demand equation at s^* becomes a nonparametric IV regression in prices. Thus micro data can cut the number of required instruments roughly in half.

5 Figures and section-by-section reading map

5.1 Figure 1: choice regions in a two-good random utility model

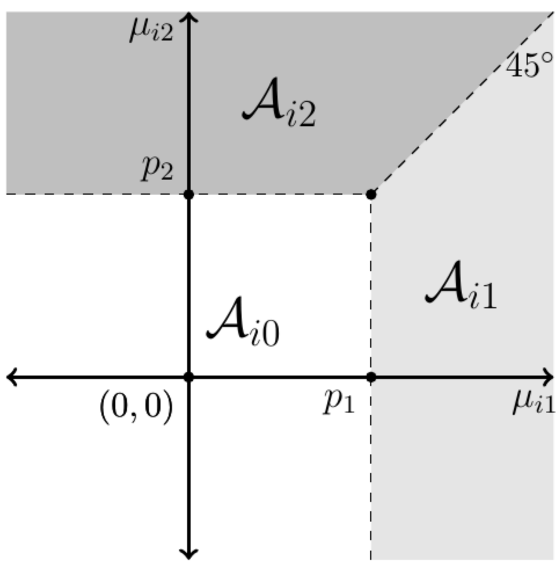

Figure 1: Choice regions for goods 0, 1, and 2.

Read: The figure partitions (μ_{i1}, μ_{i2})-space into regions where the consumer chooses the outside good, good 1, or good 2.

Economic message. Random utility demand translates heterogeneity in latent valuations into market shares. Prices shift the boundaries between regions, and substitution patterns depend on the distribution of consumers around those boundaries.

5.2 Figure 2: connected substitutes

Figure 2: Substitution in Standard Discrete Choice Models

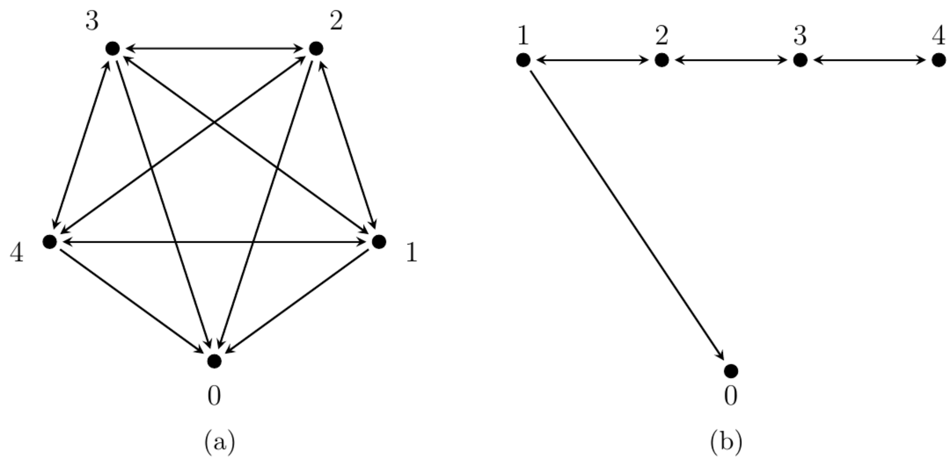

Figure 2: Substitution graphs in standard discrete choice models.

Read: The figure shows directed graphs of substitution among four inside goods and an outside good. The key property is that every inside good is connected by a substitution path to the outside good.

Economic message. Connected substitutes ensure that the demand system can be inverted. Invertibility is what allows the econometrician to recover latent demand indices from observed shares.

5.3 Sections 1–2: motivation and challenges

Read: The chapter first explains why demand elasticities are necessary for IO and policy counterfactuals. It then shows that endogeneity and multiple demand shocks make demand estimation different from regression.

Economic message. The most important conceptual point is that demand estimation is about ceteris paribus slopes at fixed latent demand states, not about average observed movements in prices and quantities.

5.4 Section 3: discrete choice demand

Read: This section introduces random utility, the outside option, mean utility, random coefficients, and why random coefficients are needed to avoid unrealistic substitution restrictions.

Economic message. Random coefficients are not just a technical complication. They are how the model lets similar goods be close substitutes and allows elasticities to depend on product characteristics, not just market shares.

5.5 Section 4: market-level BLP estimation

Read: This section turns the random-coefficient discrete choice model into an estimable GMM problem. The central algorithm repeatedly inverts market shares to recover ξ_{jt} and checks orthogonality with instruments.

Economic message. Market-level demand estimation is possible because the model transforms unobserved quality into a product-market residual that can be disciplined by instruments.

5.6 Section 5: nonparametric market-level identification

Read: The paper shows that the BLP logic is not purely parametric. With an index restriction, connected substitutes, and sufficiently relevant instruments, demand is nonparametrically identified from market-level data.

Economic message. Parametric BLP should be understood as a practical finite-sample implementation of a deeper identification logic: index, inversion, and instruments.

5.7 Sections 6–7: micro data and nonparametric micro identification

Read: Micro data reveal the joint distribution of consumer characteristics and choices. This provides within-market variation that is unavailable in aggregate market shares.

Economic message. Micro data weaken the need for BLP-style aggregate instruments, but do not eliminate the price endogeneity problem. Cross-market price instruments still matter.

5.8 Section 8: future work

Read: The paper highlights dynamics, functional-form flexibility, and invertibility as important open areas.

Economic message. The field has made large progress on identification and computation, but difficult demand settings remain: dynamic choices, complementarity, multiple purchases, and counterfactuals depending on higher-order derivatives.

6 Short summary

- Demand estimation is central because elasticities and substitution patterns determine markups, welfare, pass-through, and counterfactual equilibrium outcomes.
- Prices are endogenous because firms set them in response to demand and cost shocks.
- In differentiated-products markets, each product's demand depends on a vector of demand shocks, so demand is not a scalar-error regression.
- Randomizing or instrumenting prices alone generally identifies only average responses, not the demand slopes needed for counterfactuals.

- Random utility models provide a tractable way to model many goods and consumer heterogeneity.
- Simple logit models are too restrictive because shares largely determine substitution patterns.
- Random coefficients make substitution patterns more realistic by linking consumer tastes to product characteristics.
- BLP estimation uses the inversion from shares to mean utilities, then imposes orthogonality between recovered demand shocks and instruments.
- Market-level nonparametric identification requires an index restriction, invertibility through connected substitutes, and instruments for both prices and shares.
- BLP instruments matter because rival characteristics can shift market shares at fixed prices.
- Supply-side moments can improve precision and help discriminate among conduct assumptions.
- Micro data help because within-market demographic variation shifts choices while holding product-market demand shocks fixed.
- Consumer panels, ranked choices, and hybrid data add information about substitution and heterogeneity.
- The broad empirical lesson is to match the data source, instrument strategy, and functional-form restrictions to the counterfactual object of interest.

7 Conclusion

Berry and Haile's chapter is best read as a guide to the logic behind modern demand estimation. Its central lesson is that the complexity of BLP-style demand estimation is not accidental. It arises because economists want demand slopes and counterfactual responses at fixed latent demand states in a multi-good equilibrium environment.

The chapter's most useful organizing principle is **index–inversion–instruments**. The index creates a tractable way to summarize unobserved demand quality. Inversion recovers those latent indices from observed shares. Instruments then separate exogenous variation in prices and quantities from endogenous variation caused by demand shocks.

The paper also clarifies why better data matter. Market-level data can identify demand, but require strong instrument variation. Micro data, panels, ranked choices, and hybrid data add within-market and within-consumer variation that directly informs preference heterogeneity and substitution. The final takeaway is therefore practical: demand estimation is credible only when the model, instruments, and data jointly provide the variation needed to hold fixed the right unobservables while moving the right economic variables.